Math340S23-FinalExam

Friday, May 5, 2023 11:37 AM

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Math 340: Elementary Matrix and Linear Algebra
FINAL EXAM

Monday, May 8, 2023, 12:25pm-2:25pm

Please circle your instructor and your Discussion Session (start time given) in the table below.

Dr. Ruhui Jin	Dr. Yousheng Shi	Dr. Kaitlyn Phillipson	Dr. Jiuhua Hu
Marios, T 7:45 AM	Qi, M 8:50 AM	Joey, M 12:05 PM	Jingyi, T 11:00 AM
Ang, T 8:50 AM	Qi, M 9:55 AM	Sun Woo, M 1:20 PM	Jingyi, R 11:00 AM
Marios, R 7:45 AM	Zaidan, W 7:45 AM	Joey, W 12:05 PM	Gautam, T 12:05 PM
Ang, R 8:50 AM	Zaidan, W 8:50 AM	Sun Woo, W 1:20 PM	Gautam, R 12:05 PM
Ang, R 9:55 AM	Qi, W 9:55 AM		Jingyi, T 9:55 AM
Emma, T 11:00 AM			
Emma, T 12:05 PM			
Marios, R 11:00 AM			
Emma, R 12:05 PM			

Name (Printed): Key ______________________ Wisc email: ______________________

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ______________________________________

READ THE FOLLOWING INFORMATION. **DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.**

- This is a 120-minute exam. It consists of 12 problems for a total of 100 points; the exam is 8 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices, P_d is the vector space of polynomials of degree at most d, $\mathbf{0}$ denotes the zero vector in a vector space V. $\mathbb{R}^n$ uses the standard vector addition/scalar multiplication unless otherwise noted.

1. (12 points, 2 points each) Are each of the following true or false statements? Circle your answer. Justification is not necessary.

a.) **T** **F** If $\lambda = 0$ is an eigenvalue of A, then A is not invertible.

b.) **T** **F** Any set of four vectors in P_2 will be linearly dependent.

c.) **T** **F** It is possible to build an injective (one-to-one) linear transformation from $\mathbb{R}^5$ to M_{22}.

$5>4 \rightarrow$ must have dim kernel ≥ 1.

d.) **T** **F** If A and B are similar matrices, then they will have the same eigenvalues and eigenvectors.

$\hookrightarrow$ may be different!!

e.) **T** **F** If A, B are 3×3 invertible matrices, then the rank of AB is at most 2.

f.) **T** **F** If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ are linearly independent, then so is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_3\}$.

$c_1\vec{v}_1 + c_2\vec{v}_2 + c_3(\vec{v}_1 + \vec{v}_3) = \vec{0}$

$(c_1 + c_3)\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$

$\Downarrow$ b/c $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are lin. indep.

$c_3 = 0,\ c_2 = 0 \Rightarrow\ c_1 + \cancel{c_3}^{0} = 0$

$c_1 = 0.$

2. (9 points total)

Suppose A, B, C are 3×3 matrices with $\det(A) = -1$, $\det(B) = 6$, and $\det(C) = 0$.

a. (3 points) Find $\det(2AB^{-1})$. Show all work.

$$\det(2AB^{-1}) = 2^3 \det(A) \cdot \frac{1}{\det(B)} = 8 \cdot -1 \cdot \frac{1}{6} = \boxed{-\frac{8}{6}}$$

b. (2 points) Is AC invertible? Briefly justify your answer.

$$\det(AC) = \det(A)\det(C) = -1 \cdot 0 = 0$$

No, b/c $\det(AC) = 0$

c. (4 points) Suppose M is an $n \times n$ invertible matrix where $M^{-1} = M^T$. Determine the possible values for the determinant of M (hint: there are two). Fully justify your answer.

$$M^{-1} = M^T$$

$$\Downarrow$$

$$\frac{1}{\det(M)} = \det(M^{-1}) = \det(M^T) = \det(M)$$

so $1 = \det(M)\det(M) = (\det(M))^2$

$$\Rightarrow \boxed{\det(M) = \pm 1}$$

3. (11 points total) Let $L : P_1 \to P_1$ be a linear transformation that is represented by the matrix

$$A = \begin{bmatrix} -4 & 1 \\ -1 & 2 \end{bmatrix}$$

with respect to the basis $S = (t+2, t-1)$.
(NOTE: S is NOT the standard basis for P_1).

a. (3 points) Compute $L(t+2)$.

$$\begin{bmatrix} -4 & 1 \\ -1 & 2 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} -4 \\ -1 \end{bmatrix}}_{=[L(t+2)]_S}$$

$$L(t+2) = -4(t+2) - 1(t-1)$$
$$= \boxed{-5t-7}$$

b. (4 points) Compute $L(5t+1)$.

$$5t+1 = a(t+2) + b(t-1) \Rightarrow \begin{aligned} a+b&=5 \\ +\ 2a-b&=1 \\ \hline 3a&=6 \Rightarrow a=2 \Rightarrow b=3. \end{aligned}$$

$$\begin{bmatrix} -4 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -8+3 \\ -2+6 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \end{bmatrix}$$

$$-5(t+2) + 4(t-1) = \boxed{-t-14}$$

c. (4 points) Is L invertible? Justify your answer.

$$\det\begin{bmatrix} -4 & 1 \\ -1 & 2 \end{bmatrix} = -8+1 = -7 \neq 0.$$

Yes! b/c A is invertible, L is invertible.

4. (5 points) Consider the subset W of M_{22} of matrices of the form $\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ where $a_1a_2a_3a_4 \geq 0$, in other words,

$$W = \left\{ \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} : a_1a_2a_3a_4 \geq 0 \right\}$$

Show through explicit counterexample(s) that W is NOT a subspace of M_{22}.

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$1 \cdot 1 \cdot 1 \cdot 0 = 0 \downarrow$ in W.

$0 \cdot 0 \cdot 0 \cdot (-1) = 0 \downarrow$ in W.

$\hookrightarrow 1 \cdot 1 \cdot 1 \cdot (-1) = -1$. X. not in W.

W is <u>not</u> closed under addition $\Rightarrow$ W is not a subspace.

5. (5 points) Consider the vector space M_{32}. Which, if any, of the following vector spaces are **isomorphic** to M_{32}?

$$P_5, P_6,\ P_7,\ \mathbb{R}^4,\ M_{23}$$

Justify your answer.

$\dim(P_5) = 6$
$\dim(P_6) = 7$
$\dim(P_7) = 8$
$\dim(\mathbb{R}^4) = 4$
$\dim(M_{23}) = 6$

P_5 and M_{23} b/c they have dimension 6, same as M_{32}.

6. (7 points) Suppose $A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$ and $C^{-1} = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$. If B is a 2×2 matrix with

C^{-1} $C(A+B) = C^{-1}\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, find B. Show all work.

$$A+B = \overbrace{\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}}^{C^{-1}}\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ 15 & 11 \end{bmatrix}$$

$$B = \begin{bmatrix} 6 & 4 \\ 15 & 11 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 19 & 9 \end{bmatrix}$$

7. (10 points total) (This problem has overflow room on next page if needed)

For this problem, **use** A **and the reduced row echelon form of** A**,** $RREF(A)$, to answer the following questions.

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 0 & -1 & -5 \\ 0 & 0 & 2 & 1 & 7 \\ 1 & 2 & 1 & -3 & -2 \\ -1 & -2 & 1 & 0 & 5 \end{bmatrix}, \quad RREF(A) = \begin{array}{c} \begin{array}{ccccc} & \text{free} & & & \end{array} \\ \begin{bmatrix} 1 & 2 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}.$$

Answer each question with a brief explanation and/or work.

a.) (2 points) What is the determinant of A?

det(A) = 0 b/c RREF(A) is <u>not</u> I_5

or RREF(A) has rows of zeros ⇒ not full rank.

(other answers possible)

b.) (2 points) Does $A\mathbf{x} = \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \\ -1 \end{bmatrix} = \vec{b}$ have a unique solution?

No, b/c A is not invertible

<u>or</u> $[A \,\vdots\, \vec{b}] \to [RREF \,\vdots\, \vec{d}]$ which has zeros in the bottom two rows.

c.) (3 points) Suppose $L : \mathbb{R}^5 \to \mathbb{R}^5$ is the linear transformation $L(\mathbf{x}) = A\mathbf{x}$ (yes, this is the same matrix A as above). Give the dimension of the kernel of L AND answer: Is L one-to-one?

dim ker L = nullity(A) = 2

L is <u>not</u> one-to-one. L has nontrivial elements in its kernel b/c dim > 0.

d.) (3 points) Suppose $L : \mathbb{R}^5 \to \mathbb{R}^5$ is the linear transformation $L(\mathbf{x}) = A\mathbf{x}$ (yes, this is the same matrix A as above and in part (c)). Give a basis for the range of L AND answer: Is L onto?

Pick up the pivots! $\begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ -3 \\ 0 \end{bmatrix}$

L is <u>not</u> onto. Its range is dim 3, and $\mathbb{R}^5$ has dimension 5.

(Overflow room for Problem 7, if needed):

8. (8 points) Consider $\mathbb{R}^2$ with the standard inner product (also known as the dot product), and consider the set W of vectors which are orthogonal to $\begin{bmatrix}2\\3\end{bmatrix}$. Show W is a subspace of $\mathbb{R}^2$.

$\begin{bmatrix}a\\b\end{bmatrix}$ is in W if $\begin{bmatrix}a\\b\end{bmatrix}\cdot\begin{bmatrix}2\\3\end{bmatrix} = 2a+3b=0.$

nonempty: $\begin{bmatrix}0\\0\end{bmatrix}\cdot\begin{bmatrix}2\\3\end{bmatrix} = 0$, so $\begin{bmatrix}0\\0\end{bmatrix}$ is in W.

closed under add: let $\begin{bmatrix}a\\b\end{bmatrix}, \begin{bmatrix}c\\d\end{bmatrix}$ be in $W \Rightarrow 2a+3b=0,\ 2c+3d=0.$

$$\left(\begin{bmatrix}a\\b\end{bmatrix}+\begin{bmatrix}c\\d\end{bmatrix}\right)\cdot\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}a+c\\b+d\end{bmatrix}\cdot\begin{bmatrix}2\\3\end{bmatrix}$$

$$= 2(a+c)+3(b+d)$$

$$= \underbrace{2a+3b}+\underbrace{2c+3d}$$

$$= 0+0=0 \checkmark.$$

Closed under scalar mult: let $\begin{bmatrix}a\\b\end{bmatrix}$ be in W, r a real #.

$$\left(r\begin{bmatrix}a\\b\end{bmatrix}\right)\cdot\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}ra\\rb\end{bmatrix}\cdot\begin{bmatrix}2\\3\end{bmatrix} = 2(ra)+3(rb)$$

$$= r(2a+3b) = (r)(0)$$

$$= 0. \checkmark$$

W is a subspace!

9. (7 points) Find the value(s) of c which make the following set a basis for P_2. Justify your answer.
$S = \{t^2 - 2ct + 1, t^2 + 2, 4t + 3\}$

Rewrite as: $at^2 + bt + c \rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

Make a matrix:

$$-2c(3) - 4$$

$$\det \begin{bmatrix} 1 & 1 & 0 \\ -2c & 0 & 4 \\ 1 & 2 & 3 \end{bmatrix} = 1(-8) - 1(\cancel{(-2c)(2)}) = 0$$

$$+6c+4$$

$$= -8 + \cancel{4c} = 0$$

$$6c = 4 \quad \cancel{8 = 4c} \Rightarrow \cancel{c = 2}. \quad c = 4/6 = 2/3$$

If $c \neq \cancel{2}$ $2/3$, then the set is linearly independent. Since there's 3 vectors and $\dim(P_2) = 3$, this makes them a basis.

10. (11 points total) Consider $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ in $\mathbb{R}^2$, and define the function

$$(\mathbf{u}, \mathbf{v}) = 4u_1v_1 + u_2v_2 - u_2v_1 - u_1v_2.$$

a. (4 points) Verify the following inner product property for $(\mathbf{u}, \mathbf{v})$:
$(c\mathbf{u}, \mathbf{v}) = c(\mathbf{u}, \mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^2$ and scalar c.

$$\left(c\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \left(\begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = 4(cu_1)(v_1) + cu_2v_2 - cu_2v_1 - cu_1v_2$$
$$= c(4u_1v_1 + u_2v_2 - u_2v_1 - u_1v_2)$$
$$= c\left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) \checkmark$$

For parts (b) and (c), assume the remaining inner product properties are true for
$(\mathbf{u}, \mathbf{v}) = 4u_1v_1 + u_2v_2 - u_2v_1 - u_1v_2$.

b. (3 points) Using this inner product $(\mathbf{u}, \mathbf{v})$, find a unit vector in the same direction as $\mathbf{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

$$\|\vec{u}\| = \sqrt{(\vec{u}, \vec{u})} = \sqrt{4(-2)(-2) + (1)(1) - (1)(-2) - (-2)(1)}$$
$$= \sqrt{16 + 1 + 2 + 2} = \sqrt{21}.$$

unit vec: $\frac{1}{\sqrt{21}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

c. (4 points) Still using this inner product $(\mathbf{u}, \mathbf{v})$, determine for which value(s) of a the vector $\begin{bmatrix} a \\ 2 \end{bmatrix}$ is orthogonal to $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

$$\left(\begin{bmatrix} a \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix}\right) = 4(a)(3) + (2)(1) - (2)(3) - (a)(1) = 0$$
$$= 12a + 2 - 6 - a = 0$$
$$11a - 4 = 0$$
$$\boxed{a = 4/11}$$

11. (7 points total) Consider $A = \begin{bmatrix} 4 & -3 \\ -2 & 3 \end{bmatrix}$.

a. (5 points) Find the eigenvalues of A.

$$\det(\lambda I - A) = \det\begin{bmatrix} \lambda-4 & 3 \\ 2 & \lambda-3 \end{bmatrix} = (\lambda-4)(\lambda-3) - 6$$
$$= \lambda^2 - 7\lambda + 12 - 6$$
$$= \lambda^2 - 7\lambda + 6$$
$$= (\lambda-6)(\lambda-1)$$
$$\boxed{\lambda = 1, 6}$$

b. (2 points) Is A diagonalizable? (Hint: you do **not** need to calculate the eigenvectors to solve this problem). Briefly justify your answer.

Yes!! A is 2x2 w/ $\underline{\underline{2}}$ distinct e-values.

12. (8 points total) For $A = \begin{bmatrix} 5 & 0 & 3 \\ -9 & 2 & -9 \\ -6 & 0 & -4 \end{bmatrix}$:

a.) (7 points) Find a basis for the eigenspace associated to the eigenvalue $\lambda = 2$. Show all work.

$$2I - A = \begin{bmatrix} 2-5 & 0 & -3 \\ 9 & 2-2 & 9 \\ 6 & 0 & 2+4 \end{bmatrix} = \begin{bmatrix} -3 & 0 & -3 \\ 9 & 0 & 9 \\ 6 & 0 & 6 \end{bmatrix}$$

$$\Big\downarrow \; -\tfrac{1}{3}r_1 \to r_1 \text{ then } -9r_1 + r_2 \to r_2,\; -6r_1 + r_3 \to r_3$$

$$\begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array} \quad \begin{array}{l} b = r \\ c = s \\ a + c = 0 \Rightarrow a = -s. \end{array}$$

$$\begin{bmatrix} -s \\ r \\ s \end{bmatrix} = s \underbrace{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}} + r \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}$$

Basis!

b.) (1 point) What is the **dimension** of the eigenspace associated to the eigenvalue $\lambda = 2$? No justification necessary.

$$\boxed{2}$$

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